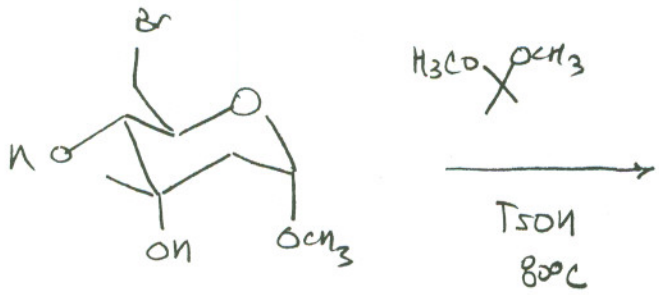
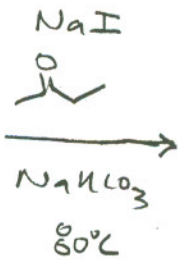
Organic Cumulative Exam October 19, 2001

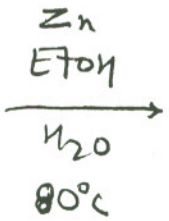
The following problems are from Baldwin and Debenham, *Organic Letters*, **2000**, *2*, 99-102. Fill in the structure of intermediates 1-13. You do not need to draw mechanisms.

Be sure that stereochemistry is unambiguously indicated where appropriate. Also be sure that your structures have reasonable bond angles; e.g. if you draw a cyclohexane, the axial and equatorial bonds should be clearly distinguishable.

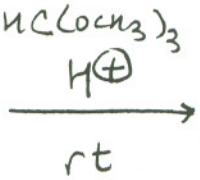

Br
H
O
OH
OCH3
H3CO
OCH3
TsOH
80°C

1

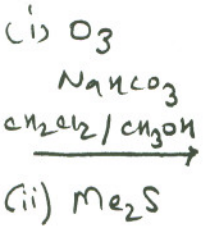

NaI
NaHCO3
80°C


Zn
EtOH
H2O
80°C

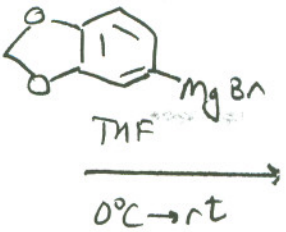
2


HC(OCH3)3
H⊕
rt

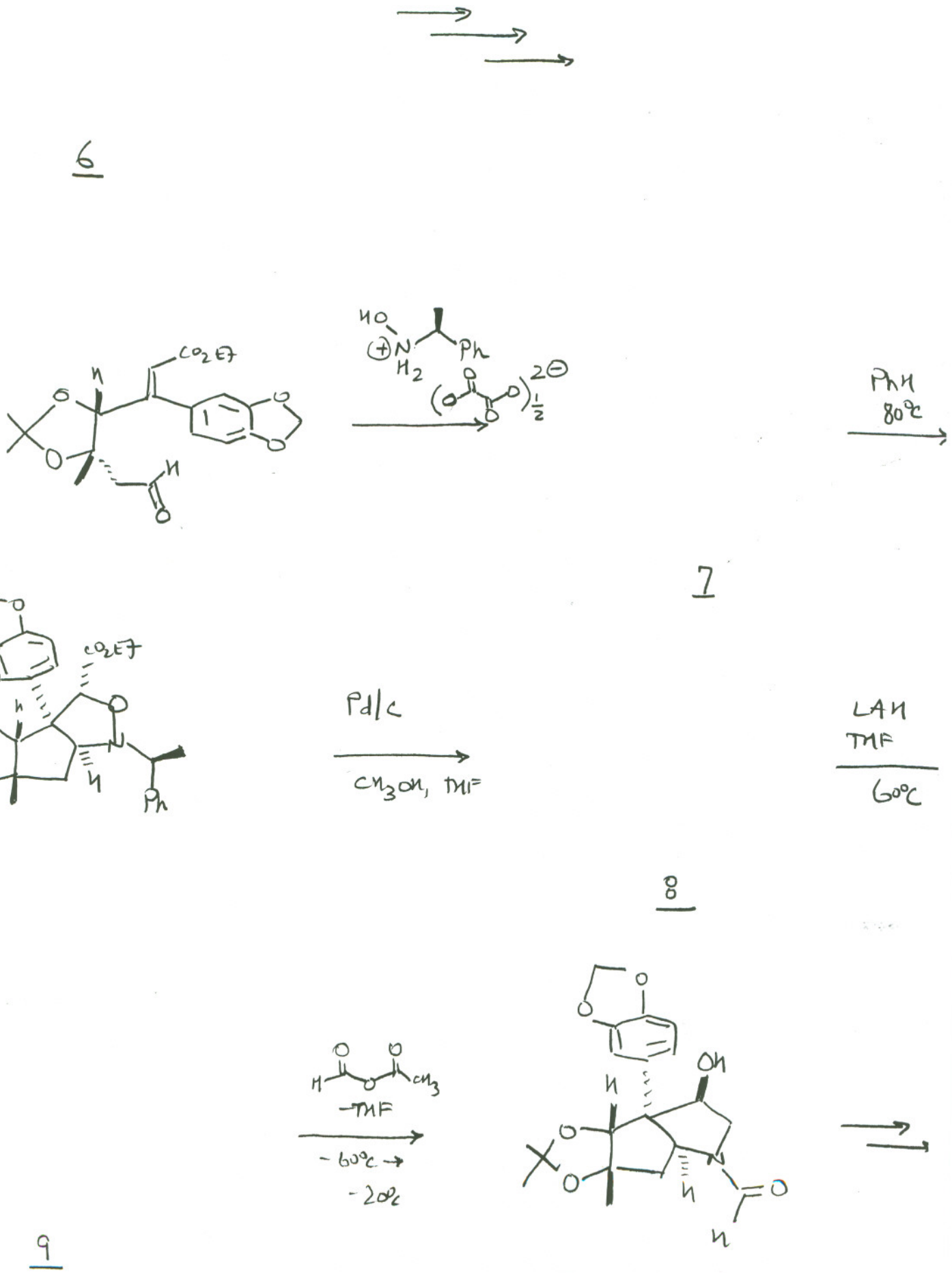
3

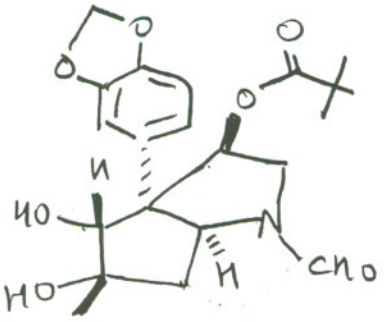

(i) O3
NaHCO3
CH2Cl2 / CH3OH
(ii) Me2S

4


MgBr
THF
0°C → rt

5


6
7
8
9
PhH
80°C
Pd/C
CH3OH, THF
LAH
THF
60°C
-THF
-60°C →
-20°C



$NaIO_4$
2:1
THF:H_2O
rt

HOAc,
PhH, rt

10

NaBH4
$CeCl_3$
CH_3OH
-78°C

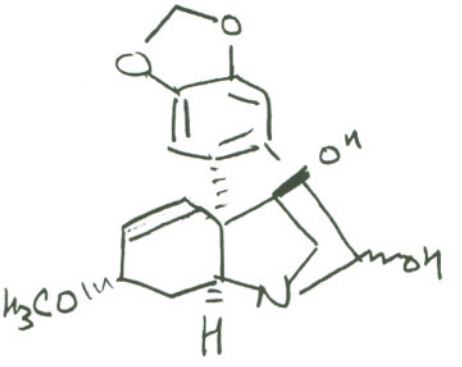
11

(i) $(MeSO_2)_2O$,
NEt_3
(ii) CH_3OH
-78°C -> 0°C

12

$POCl_3$

LiOH
CH_3OH
rt



Haemanthidine

13